TELECOM AND AI

THE CONVERGENCE OF INTELLIGENT NETWORKS AND ARTIFICIAL INTELLIGENCE

GRADUATION PROJECTS

OMAR AHMED


IEEE
Advancing Technology
for Humanity
AI
BUILDING
INTELLIGENT
SOLUTIONS
streamlit_app.py
import streamlit as st
import numpy as np
st.title("AI Network Analyzer")
st.subheader("Visualizing
Intelligent Networks")
data = np.random.rand(100, 100)
st.line_chart(data)
st.write("Building the future
with AI.")
numpy_example.py
import numpy as np
x = np.linspace(0, 10, 100)
y = np.sin(x)
z = np.cos(x)
result = np.vstack((x, y, z)).T
NETWORK PERFORMANCE
NEURAL
NETWORKS
ROBOTICS
DEEP
LEARNING
Mariem Mohie
PM
AI
Mariem Mohie
MEMBER AT IEEE SSCS AUSC AI TEAM

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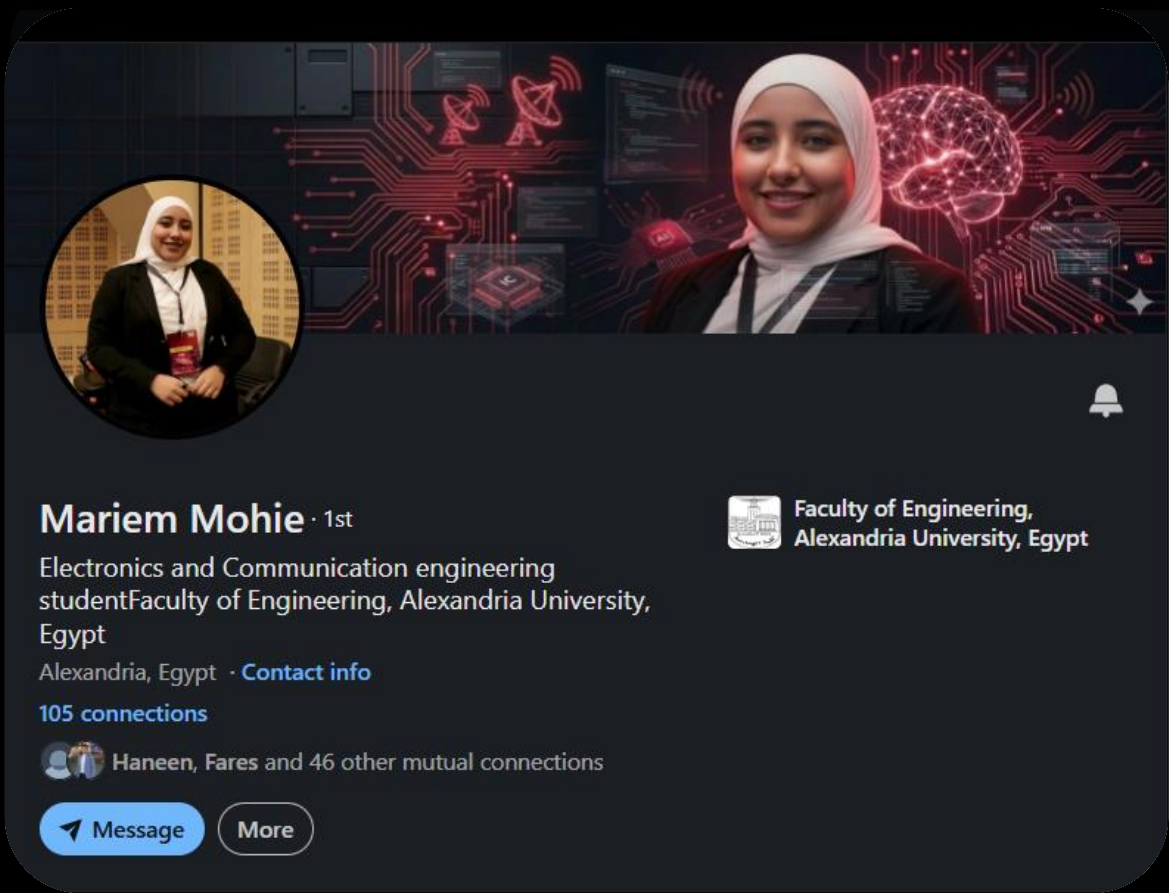

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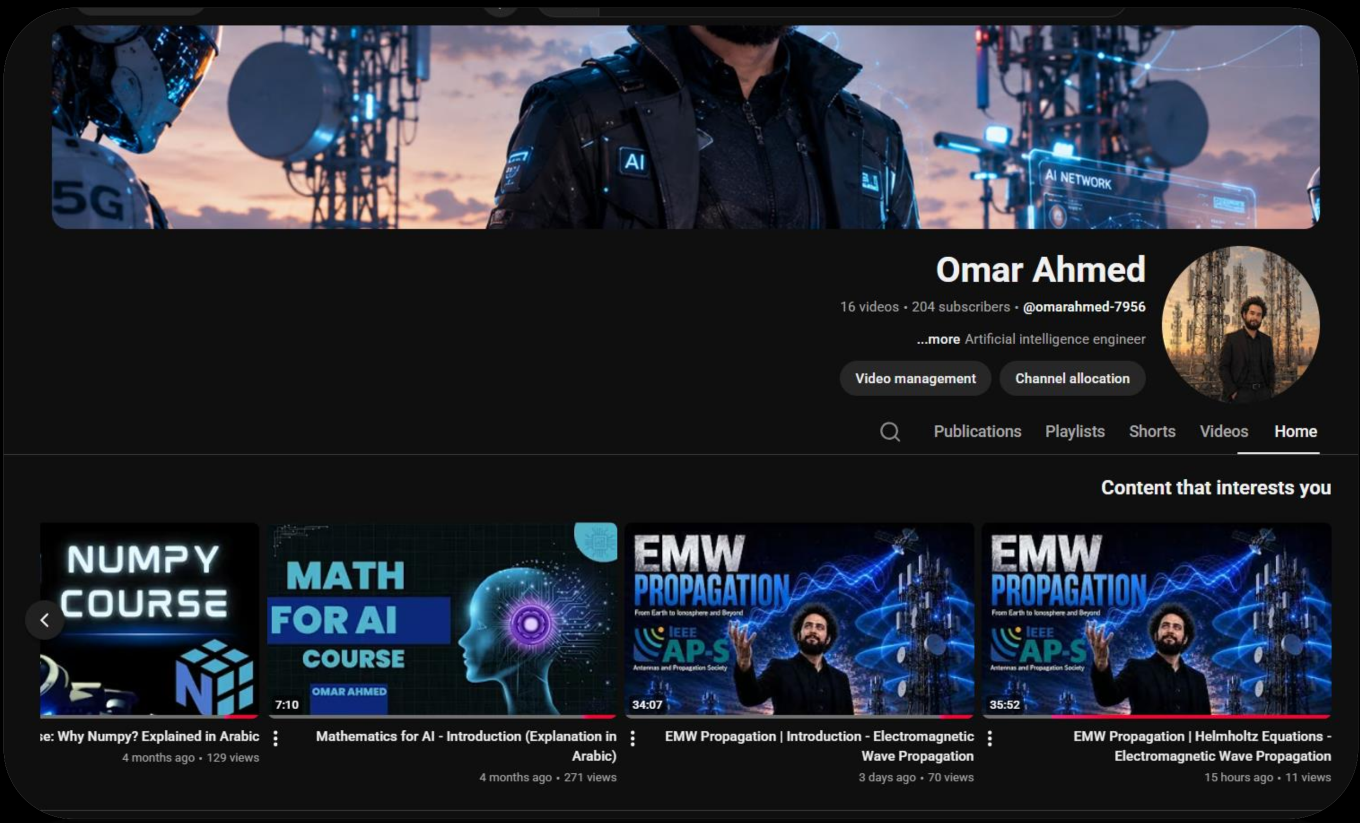



POWER CONTROL
OPTIMIZATION
USING AI
INTELLIGENT POWER. OPTIMIZED NETWORKS.
AI
POWER CONTROL GOAL
SINR
Optimized
Non-Optimized
−10
0
10
20
30
Transmit Power (dBm)
OPTIMIZATION WITH AI
NETWORK STATE
AI MODEL
POWER ALLOCATION
OPTIMIZED PERFORMANCE
OPTIMIZATION PROBLEM
$\underset{p}{\text{maximize}} \quad \sum_{i=1}^{N} w_i \log_2 (1 + SINR_i)$
$\text{subject to} \quad 0 \leq p_i \leq P_i^{max}, \ \forall i$
$\sum_{i=1}^{N} p_i \leq P_{total}$
AI MODEL ARCHITECTURE
INPUT
(Network State)
HIDDEN
LAYERS
OUTPUT
(Power Allocation)
OBJECTIVE
Optimize Transmit Power
Maximize Throughput
Minimize Interference
Improve Energy Efficiency
NETWORK PERFORMANCE
92%
THROUGHPUT
-78%
INTERFERENCE
AI

The goal is to choose the best transmit power for each user/device/base station so that the network gets high SINR, high throughput, low interference, and low energy consumption.

Traditional power control uses fixed mathematical rules. But real telecom networks change every second because of:

mobility, fading, shadowing, user density, interference, traffic load, and handover.

AI can learn from the environment and adapt dynamically. Reinforcement learning is especially suitable because power control is a decision-making problem. Surveys on 5G/6G ML mention RL as useful for resource allocation, beamforming, modulation, and power control

Best project formulation

<u>Use Reinforcement Learning</u>.

Think of the telecom network as an environment.

State

The input to the AI agent can be:

$$S_t = [SINR, RSSI\ , RSRP, RSRQ, interference, distance, \\ channel\ gain, current\ power\ ]$$

Example :

User distance = 300m

Channel gain = -90dB

Current transmit power = 18 dBm

Interference = -95 dBm

SINR = 8 dB

The AI Chooses the new transmit power :

- Decrease power
- Keep power
- Increase power

Model you can use like Q-Learning , Deep Q-Network (DQN), DDPG / PPO / SAC

I recommend start simple :

Phase 1 : Traditional algorithm baseline

Phase 2 : Q-Learning

Phase 3 : DQN

Phase 4 : compare results

Project should produce graphs like (SINR vs Time – Transmit power vs Time – Throughput vs Time – Energy efficiency vs time – Interference vs time – Reward Curve – AI vs Traditional methods)

$$Energy\ Efficiency = \frac{Throughput}{power\ consumption}$$

$$Spectral\ Efficiency = \frac{R}{B}$$

Start with a Python simulator. Generate users randomly, calculate path loss, received power, interference, SINR, and throughput.

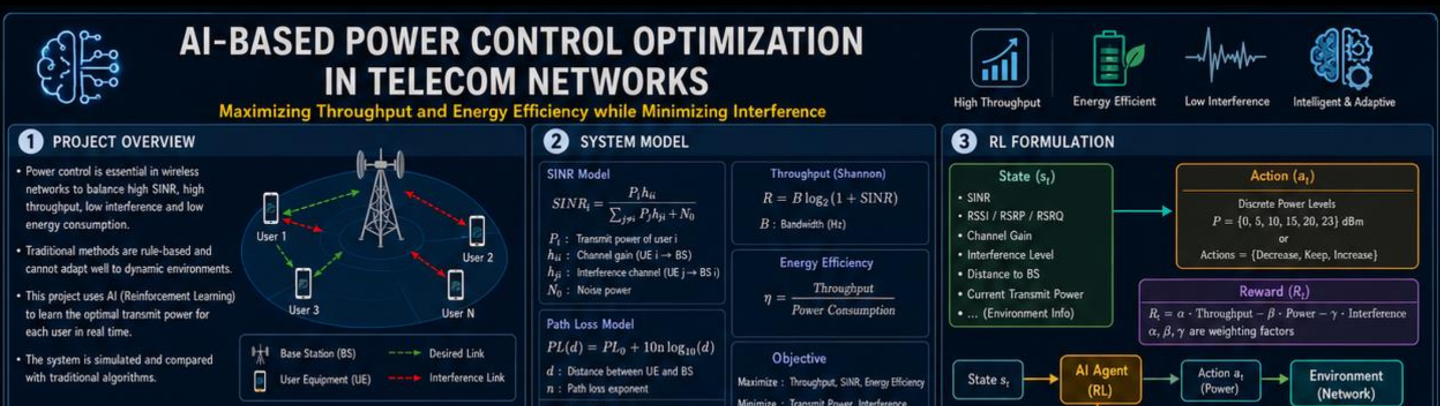
Then implement fixed power and random power as baselines.

After that, build a Q-learning or DQN agent that learns the best power level.

Finally, create app where the user can change:

- Number of users
- Cell radius
- Bandwidth
- Noise power
- Maximum transmit power
- AI algorithm

and show all graphs.




RESOURCE ALLOCATION USING ML
SMART ALLOCATION • MAXIMUM EFFICIENCY • OPTIMIZED PERFORMANCE
ML
MACHINE LEARNING MODEL
INPUT → MODEL → PREDICTION → ALLOCAT
RESOURCE ALLOCATION
SYSTEM RESOURCES
BANDWIDTH 78%
COMPUTE 62%
POWER 85%
CHANNELS 71%
Mariem Mohie
PM
ALLOCATION OPTIMIZATION
maximize $\sum_{i=1}^{N} U_i(x_i)$
subject to $\sum_{i=1}^{N} x_i \leq C$
$0 \leq x_i \leq x_i^{max}, \quad \forall i$
$U_i(x_i)$: Utility of user i
x_i : Resource allocated to user i
C : Total available resource
PERFORMANCE IMPROVEMENT
THROUGHPUT +42%
EFFICIENCY +37%
FAIRNESS +28%

How can the network distribute its limited resources intelligently among users to maximize performance?

These resources include Power – Bandwidth - Frequency channels - Time slots - RBs (Resource Blocks) -Antennas - Beamforming directions - CPU/Edge resources - Caching resources

Traditional telecom systems use mathematical optimization and rule-based schedulers.

Machine Learning tries to make the allocation:

- adaptive
- intelligent
- predictive
- self-optimizing
- real-time

Imagine a cellular tower, The tower has limited power, limited bandwidth, limited spectrum and limited antennas, But many users are connected simultaneously.

Example : 100 users, Only 20 MHz bandwidth
Limited transmit power

The network must decide:

- Who gets bandwidth?
- Who gets higher power?

- Who transmits now?
- Which modulation?
- Which beam?
- Which frequency?

This entire process is called Resource Allocation

The main problem is Wireless resources are LIMITED but demand is HUGE, This creates trade-offs.

If you give one user more resources another user gets less, interference increases, energy consumption increases, So the system must optimize globally.

Types of Resource Allocation

Power Allocation

Most famous where Goal is

Choose transmit power for users.

Optimization target:

- **maximize SINR**
- **minimize interference**
- **save energy**

Bandwidth Allocation

The BS decides How much bandwidth each user gets

Example:

Video user → larger bandwidth
IoT sensor → tiny bandwidth

Channel Allocation

Assign frequency channels to users.

Problem:

Two nearby users using same frequency create interference.

This becomes similar to graph coloring problems.

Time Slot Scheduling

Who transmits first?

Used heavily in:

- **LTE**
- **5G NR**

ML can learn better scheduling.

Beamforming Allocation

Massive MIMO systems must choose:

Which beam points to which user

Huge optimization problem.

User Association

Which base station should serve the user?

Especially important in:

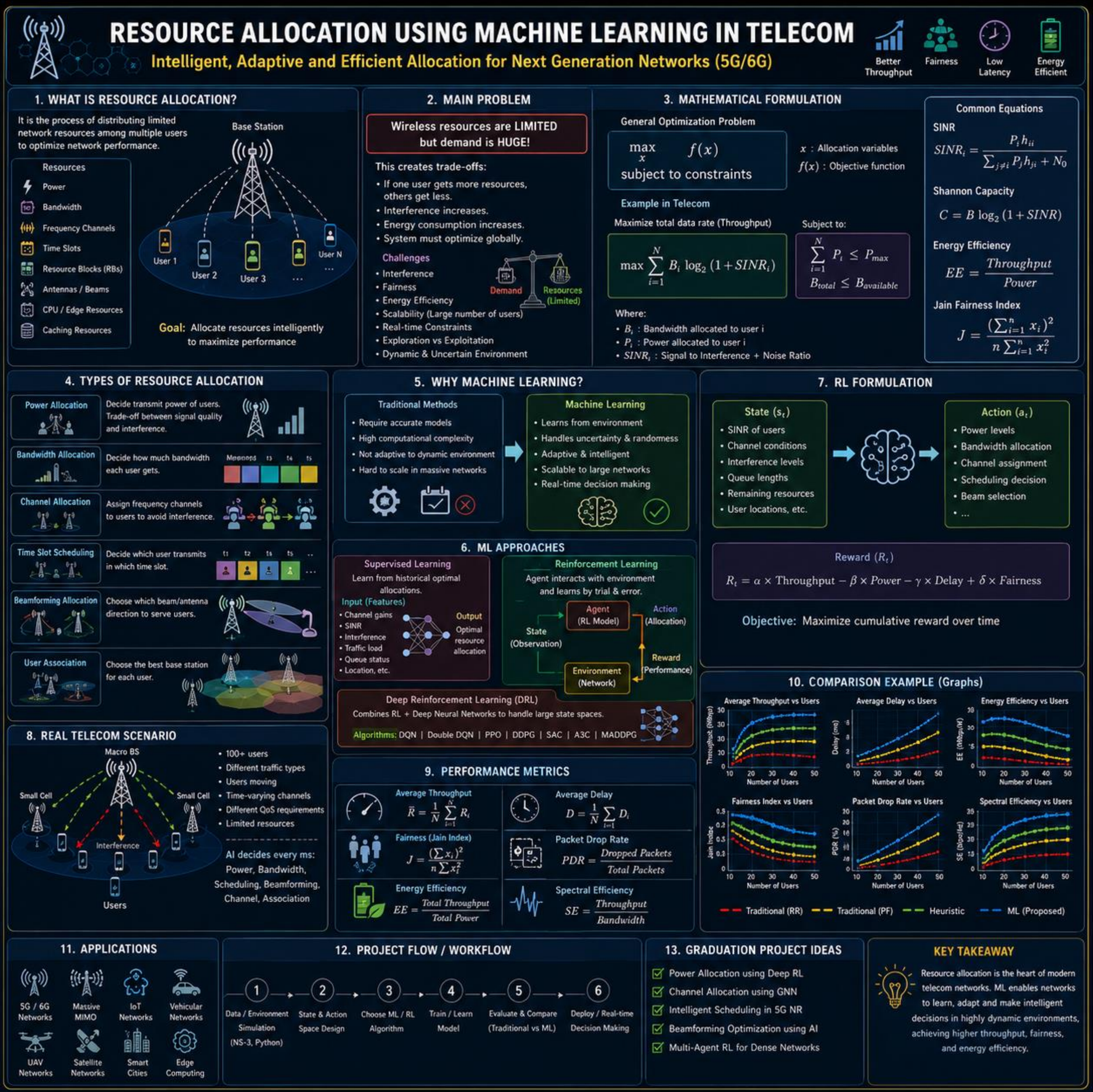
HetNets
Small cells
6G

Real Telecom Scenario

100 users + Users moving + Different traffic + Different interference

Every millisecond the BS must decide Power Bandwidth, Scheduling, Beamforming

The AI model acts like a brain controlling the network.


RESOURCE ALLOCATION USING MACHINE LEARNING IN TELECOM
Intelligent, Adaptive and Efficient Allocation for Next Generation Networks (5G/6G)
Better Throughput
Fairness
Low Latency
Energy Efficient
1. WHAT IS RESOURCE ALLOCATION?
It is the process of distributing limited network resources among multiple users to optimize network performance.
Resources
Power
Bandwidth
Frequency Channels
Time Slots
Resource Blocks (RBs)
Antennas / Beams
CPU / Edge Resources
Caching Resources
Base Station
User 1
User 2
User 3
User N
Goal: Allocate resources intelligently to maximize performance
2. MAIN PROBLEM
Wireless resources are LIMITED but demand is HUGE!
This creates trade-offs:
• If one user gets more resources, others get less.
• Interference increases.
• Energy consumption increases.
• System must optimize globally.
Challenges
• Interference
• Fairness
• Energy Efficiency
• Scalability (Large number of users)
• Real-time Constraints
• Exploration vs Exploitation
• Dynamic & Uncertain Environment
Demand
Resources (Limited)
3. MATHEMATICAL FORMULATION
General Optimization Problem
$\max_x \quad f(x)$
subject to constraints
x : Allocation variables
$f(x)$: Objective function
Example in Telecom
Maximize total data rate (Throughput)
$\max \sum_{i=1}^{N} B_i \log_2(1+SINR_i)$
Subject to:
$\sum_{i=1}^{N} P_i \leq P_{max}$
$B_{total} \leq B_{available}$
Where:
• B_i : Bandwidth allocated to user i
• P_i : Power allocated to user i
• $SINR_i$: Signal to Interference + Noise Ratio
Common Equations
SINR
$SINR_i = \frac{P_i h_{ii}}{\sum_{j \neq i} P_j h_{ji} + N_0}$
Shannon Capacity
$C = B \log_2(1+SINR)$
Energy Efficiency
$EE = \frac{Throughput}{Power}$
Jain Fairness Index
$J = \frac{(\sum_{i=1}^{n} x_i)^2}{n \sum_{i=1}^{n} x_i^2}$
4. TYPES OF RESOURCE ALLOCATION
Power Allocation
Decide transmit power of users. Trade-off between signal quality and interference.
Bandwidth Allocation
Decide how much bandwidth each user gets.
Channel Allocation
Assign frequency channels to users to avoid interference.
Time Slot Scheduling
Decide which user transmits in which time slot.
Beamforming Allocation
Choose which beam/antenna direction to serve users.
User Association
Choose the best base station for each user.
5. WHY MACHINE LEARNING?
Traditional Methods
• Require accurate models
• High computational complexity
• Not adaptive to dynamic environment
• Hard to scale in massive networks
Machine Learning
• Learns from environment
• Handles uncertainty & randomness
• Adaptive & intelligent
• Scalable to large networks
• Real-time decision making
6. ML APPROACHES
Supervised Learning
Learn from historical optimal allocations.
Input (Features)
• Channel gains
• SINR
• Interference
• Traffic load
• Queue status
• Location, etc.
Output
Optimal resource allocation
Reinforcement Learning
Agent interacts with environment and learns by trial & error.
Agent (RL Model)
Action (Allocation)
State (Observation)
Reward (Performance)
Environment (Network)
Deep Reinforcement Learning (DRL)
Combines RL + Deep Neural Networks to handle large state spaces.
Algorithms: DQN | Double DQN | PPO | DDPG | SAC | A3C | MADDPG
7. RL FORMULATION
State (s_t)
• SINR of users
• Channel conditions
• Interference levels
• Queue lengths
• Remaining resources
• User locations, etc.
Action (a_t)
• Power levels
• Bandwidth allocation
• Channel assignment
• Scheduling decision
• Beam selection
• ...
Reward (R_t)
$R_t = \alpha \times Throughput - \beta \times Power - \gamma \times Delay + \delta \times Fairness$
Objective: Maximize cumulative reward over time
8. REAL TELECOM SCENARIO
Macro BS
Small Cell
Small Cell
Interference
Users
• 100+ users
• Different traffic types
• Users moving
• Time-varying channels
• Different QoS requirements
• Limited resources
AI decides every ms: Power, Bandwidth, Scheduling, Beamforming, Channel, Association
9. PERFORMANCE METRICS
Average Throughput
$\bar{R} = \frac{1}{N} \sum_{i=1}^{N} R_i$
Average Delay
$D = \frac{1}{N} \sum_{i=1}^{N} D_i$
Fairness (Jain Index)
$J = \frac{(\sum x_i)^2}{n \sum x_i^2}$
Packet Drop Rate
$PDR = \frac{Dropped\ Packets}{Total\ Packets}$
Energy Efficiency
$EE = \frac{Total\ Throughput}{Total\ Power}$
Spectral Efficiency
$SE = \frac{Throughput}{Bandwidth}$
10. COMPARISON EXAMPLE (Graphs)
Average Throughput vs Users
Average Delay vs Users
Energy Efficiency vs Users
Fairness Index vs Users
Packet Drop Rate vs Users
Spectral Efficiency vs Users
Number of Users
Traditional (RR)
Traditional (PF)
Heuristic
ML (Proposed)
11. APPLICATIONS
5G / 6G Networks
Massive MIMO
IoT Networks
Vehicular Networks
UAV Networks
Satellite Networks
Smart Cities
Edge Computing
12. PROJECT FLOW / WORKFLOW
1 Data / Environment Simulation (NS-3, Python)
2 State & Action Space Design
3 Choose ML / RL Algorithm
4 Train / Learn Model
5 Evaluate & Compare (Traditional vs ML)
6 Deploy / Real-time Decision Making
13. GRADUATION PROJECT IDEAS
Power Allocation using Deep RL
Channel Allocation using GNN
Intelligent Scheduling in 5G NR
Beamforming Optimization using AI
Multi-Agent RL for Dense Networks
KEY TAKEAWAY
Resource allocation is the heart of modern telecom networks. ML enables networks to learn, adapt and make intelligent decisions in highly dynamic environments, achieving higher throughput, fairness, and energy efficiency.


TRAFFIC PREDICTION USING AI
SMART PREDICTION • LESS CONGESTION • BETTER MOBILITY
AI
ESTIMATED ARRIVAL TIME
12 MIN
AI PREDICTION MODEL
DATA INPUT
AI MODEL
PREDICTION
TRAFFIC FORECAST
TRAFFIC FORECAST
Low
Medium
High
TRAFFIC FEATURES
VEHICLE COUNT 86%
AVERAGE SPEED 69%
ROAD DENSITY 75%
WEATHER IMPACT 61%
TIME & DAY FACTOR 82%
Mariam Mohie
PM
PREDICTION BENEFITS
REDUCE CONGESTION -35%
IMPROVE ETA ACCURACY +40%
LOWER EMISSIONS -28%
AI ALGORITHM
DATA COLLECTION
FEATURE EXTRACTION
MODEL TRAINING
TRAFFIC PREDICTION
REAL-TIME UPDATE
PREDICTED VS ACTUAL
Predicted
Actual
750
500
250
0
00:00
06:00
12:00
18:00
24:00
REAL-TIME TRAFFIC FLOW
DENSITY
Medium
SPEED
45 km/h
FLOW
High

Can AI predict future network load and traffic patterns before they occur?

Example:

Current traffic = 300 Mbps
AI predicts:
After 10 minutes = 900 Mbps

Telecom traffic is highly dynamic and nonlinear

Traffic depends on:

- Time
- Location
- Events
- Weather
- Mobility
- Applications
- Human behavior

Traffic is NOT random only, It contains hidden patterns.

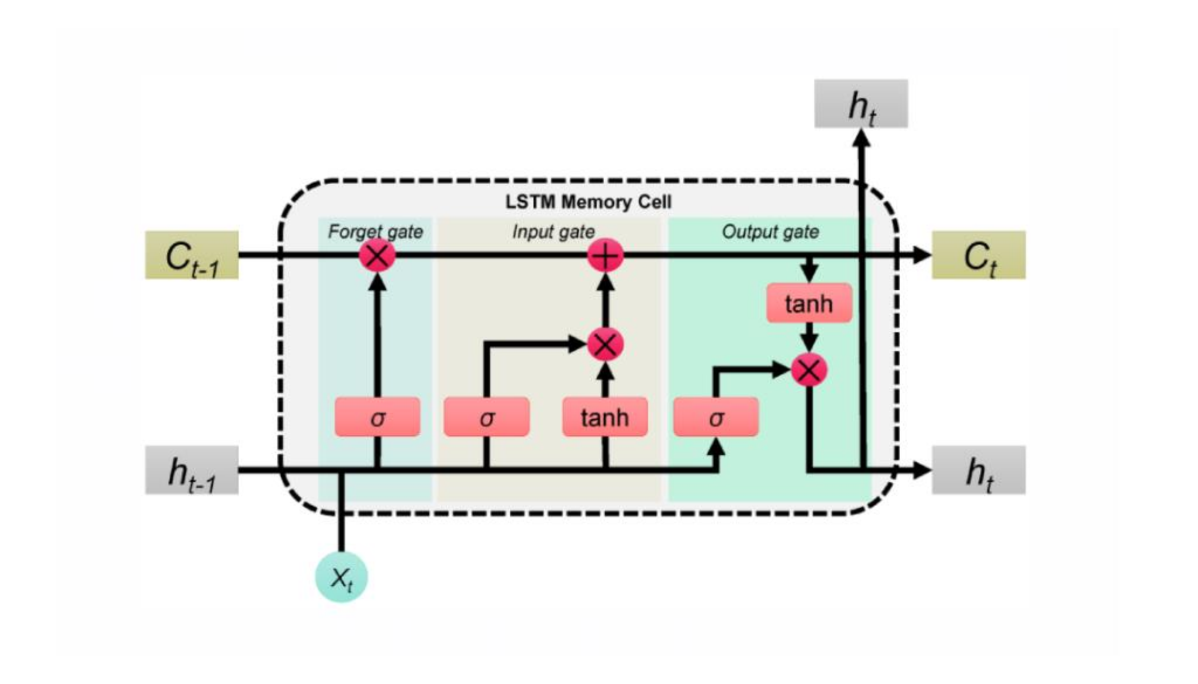
AI tries to learn these patterns.

Why is this important?

Without prediction Congestion occurs suddenly, Latency increases, Packet loss increases, QoS degrades

With prediction Resources allocated early, Load balancing activated, Power adjusted, Extra BS activated, Edge caching prepared

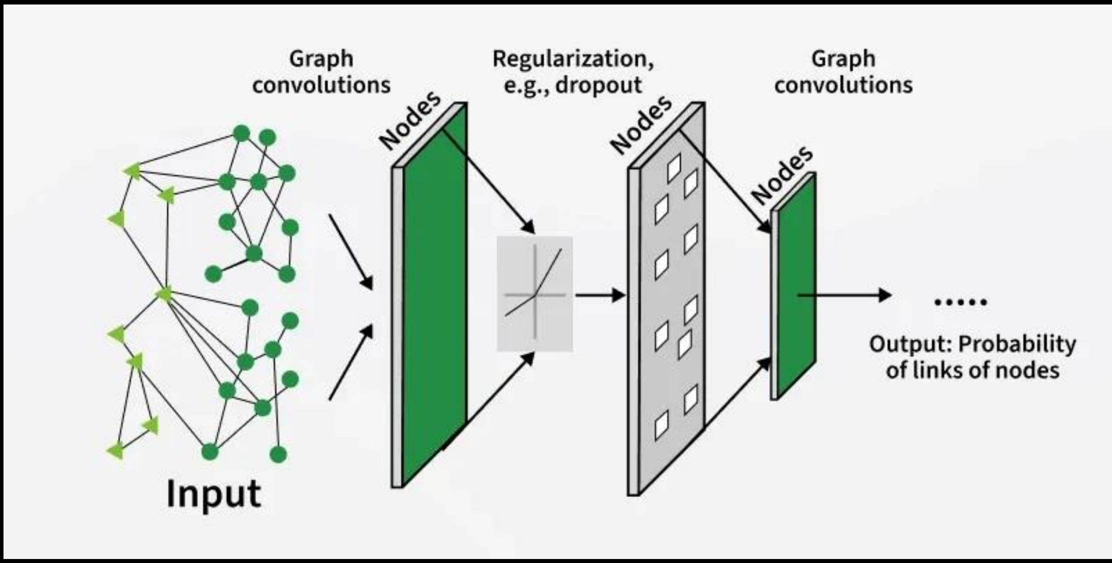
Most Important AI Models



LSTMs are Very important for sequential telecom traffic.

Because traffic is time-dependent, LSTM remembers previous patterns.

Graph Neural Networks also can good at this task



Strong Project Architecture

Phase 1 — Dataset

Collect or generate

Traffic over time, Users, Bandwidth usage, Cell load

Phase 2 — Visualization

Plot Traffic vs time, Congestion periods, Peak hours, Heatmaps

Phase 3 — AI Model

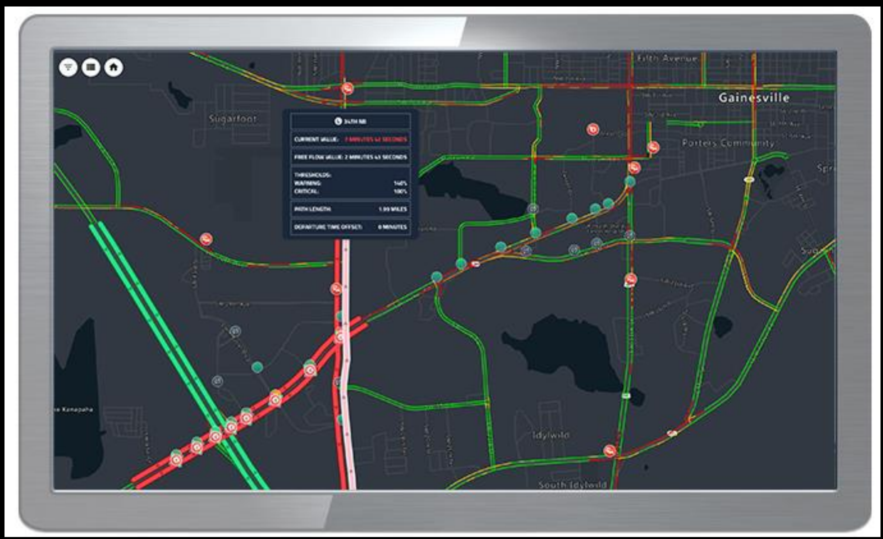
Train LSTM, GRU, Transformer

Input = previous traffic sequence.

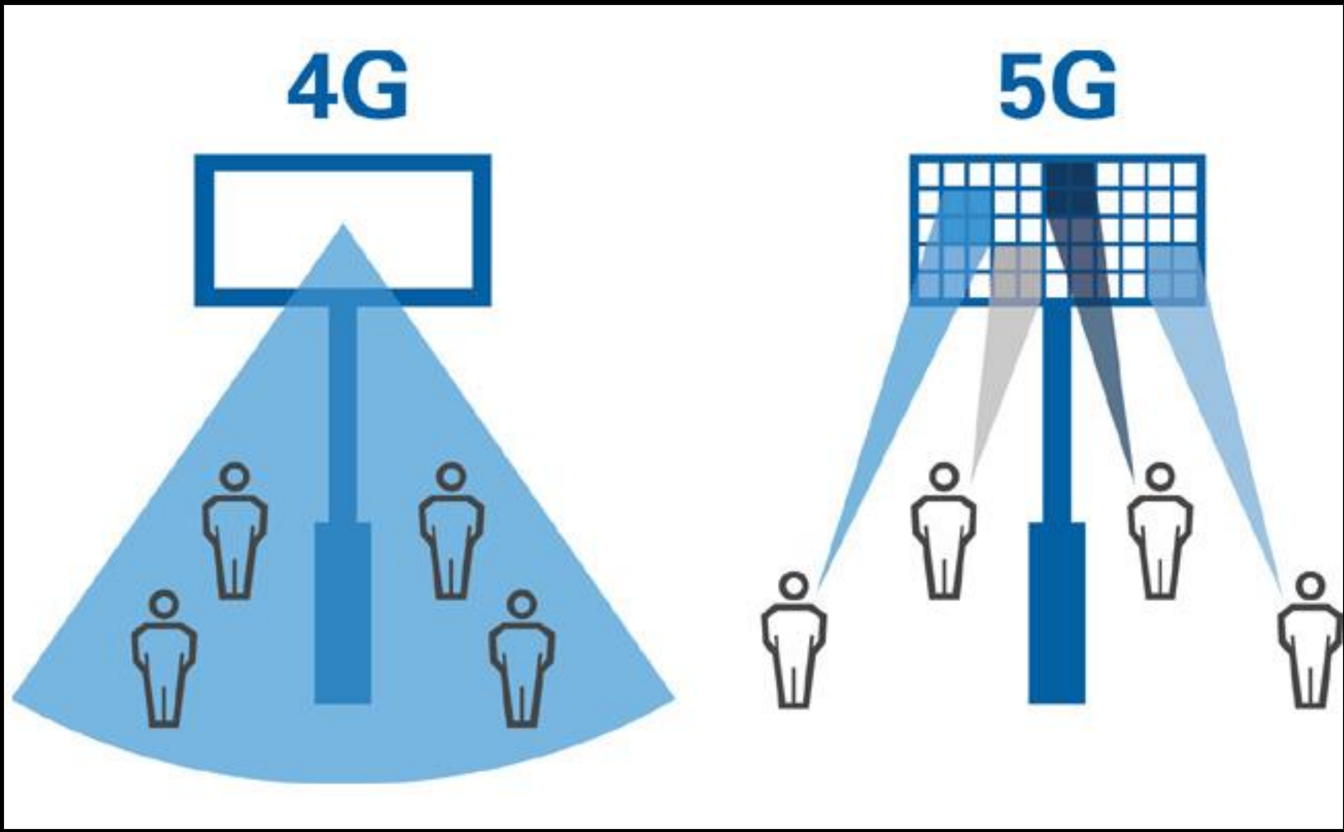
Output = future traffic.

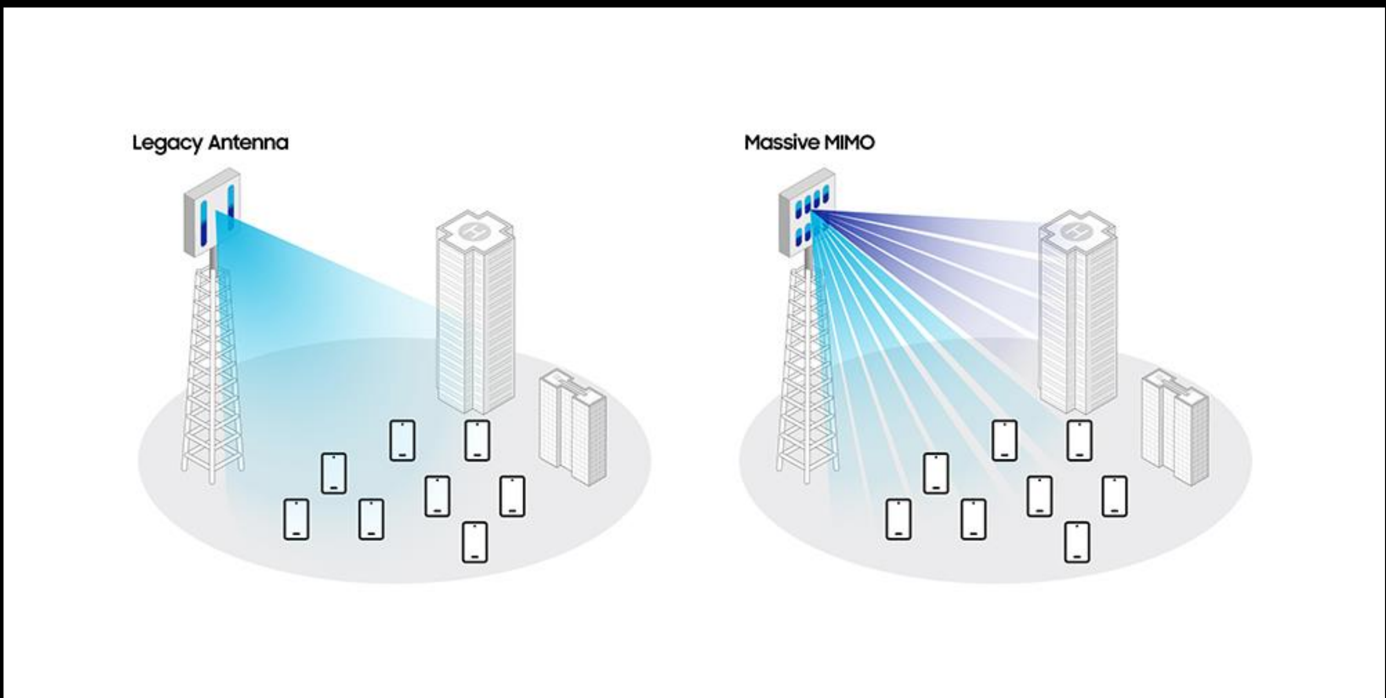
Phase 4 — Optimization Layer

After prediction use predicted traffic to Allocate bandwidth, Control power or Schedule users




MASSIVE
MIMO
PROJECTS
CONNECTING INTELLIGENCE. EMPOWERING THE FUTURE.
SECTOR 1
SECTOR 2
SECTOR 2
SECTOR 5
SECTOR 6
SECTOR 7
EMW
MASSIVE MIMO
$\Delta f = \frac{1}{T_{sym}}$
EMF EXPOSURE
$E = \sqrt{\frac{2PZ_0}{4\pi r^2}}$
BEAMFORMING
$w^* = \arg\max_w |h^H w|^2$
SPECTRAL EFFICIENCY
$\eta = \frac{B}{\log_2(1+\mathrm{SINR})}$
PATH LOSS
$PL(dB) = 10n\log_{10}(d) + X_\sigma$
CHANNEL MODEL
$H = H_{LOS} + H_{NLOS}$


4G
5G

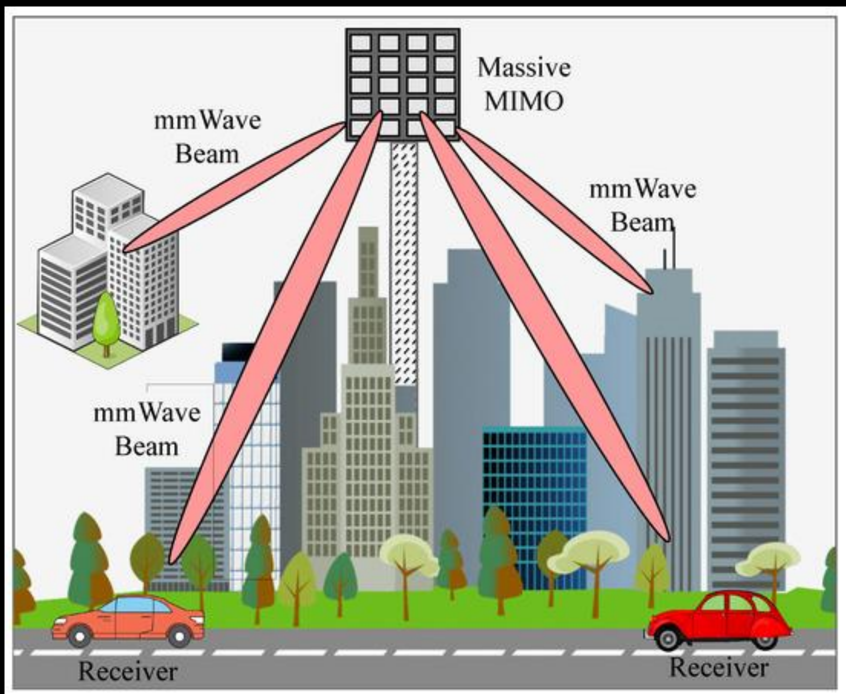

Legacy Antenna
Massive MIMO

Watch this video to know more about massive memo

https://drive.google.com/file/d/1Vj76CpfFP8TqyYuk2tZen-0FUi54Bwxl/view?usp=sharing

project 1 : AI-Based Beamforming Optimization in Massive MIMO Networks

Use AI to automatically steer antenna beams toward users in a Massive MIMO system.



The Main Problem

In Massive MIMO:

64 antennas
128 antennas
256 antennas

the base station must continuously decide:

Beam direction
Phase shifts
Power per antenna
User-beam mapping

This optimization becomes extremely complex.

Traditional beamforming algorithms like ZF, MMSE, MRT require heavy matrix computations.

AI can learn optimal beamforming dynamically.

Massive antenna arrays create narrow beams.

AI learns Which beam direction gives best SINR for every user.

project 2 : Traffic Prediction and Dynamic Resource Allocation in Massive MIMO Using AI

This is a HUGE integrated telecom intelligence project.

The idea:

Predict future network traffic and dynamically optimize Massive MIMO resources before congestion occurs.

This project combines:

- traffic prediction
- resource allocation
- beamforming
- AI scheduling
- Massive MIMO

all together.

AI predicts traffic before it happens.

Project 3 : AI-Based Channel Estimation and Interference Mitigation in Massive MIMO

Main Problem

Massive MIMO requires accurate:

Channel State Information (CSI)

But estimating CSI is difficult because of:

Noise
Fast fading
Mobility
Pilot contamination
Interference

Pilot Contamination

One of the biggest Massive MIMO problems.

Neighboring cells reuse pilot signals.

Result is Wrong channel estimation, Bad beamforming, Reduced SINR